Undecidable Languages

SELFREJECT : $\{\langle M\rangle : M \text{ rejects } \langle M\rangle\}$

SELFACCEPT : $\{\langle M\rangle : M \text{ accepts } \langle M\rangle\}$

SELFHALT ←

SELFDIVERGE

HALT : $\{\langle M, w\rangle : M \text{ halts on } w\}$

Theorem: HALT is undecidable.

Proof: Suppose not. Then let H be a TM that decides HALT.

H accepts $\langle M, w\rangle \iff M$ halts on w.

idea: use H to build a decider for some language known to be undecidable.

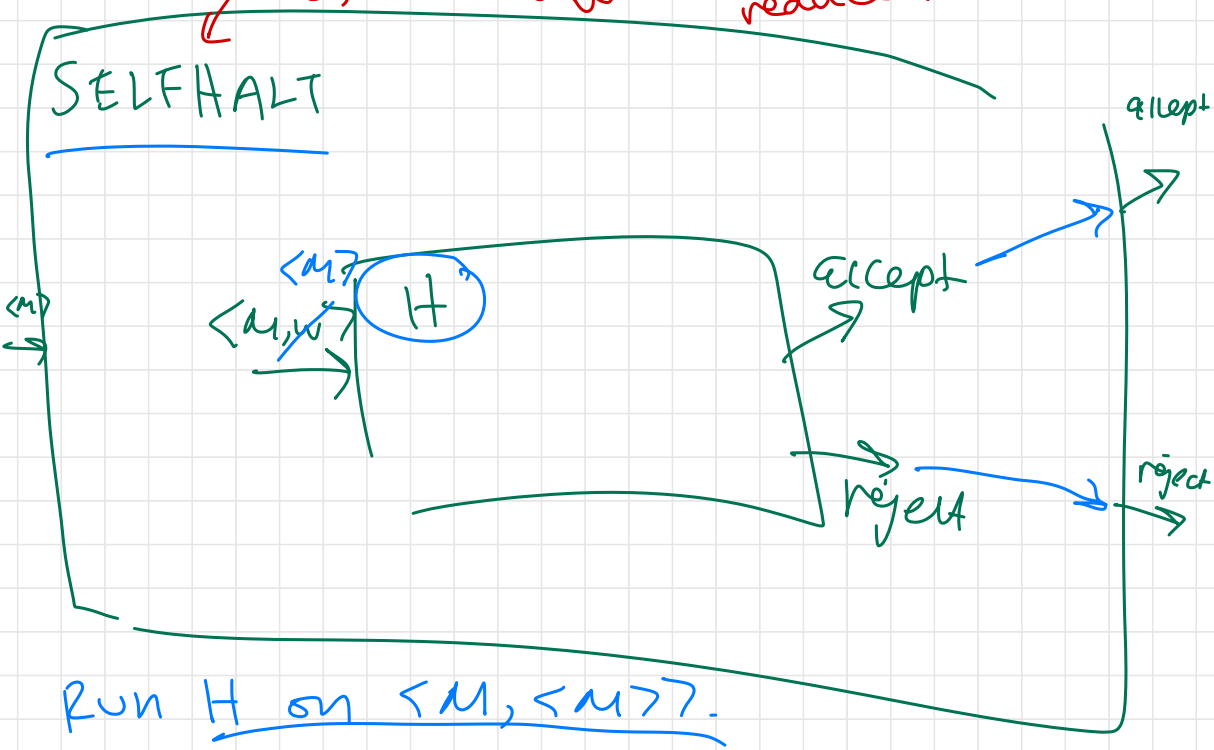
Let's try SELFHALT.

a decider for SELFHALT must:

✓ accept any $\langle M\rangle$ that halts on itself

✓ reject any $\langle M\rangle$ that does not.

are there other languages we could have reduced from?



Run H on <M, <M>>.

what happens if M halts on itself?

if M does not halt on itself?

So we can decide SELFHALT by running H on <M, <M>>.

But this is a contradiction, because SELFHALT is undecidable. □

This is a proof by reduction.

If we reduce problem A to problem B,

If we knew A was undecidable, then we know B is undecidable as well.

<u>Q</u> Is DIVERGE : $\{\langle M, w\rangle : M \text{ diverges on } w\}$ undecidable?

Would you prove by reduction?